

1. Practical 6

Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

```
sri_dhruthi@Dhruthi: ~  
GNU nano 8.7.1 practical6b.c  
#include <stdio.h>  
#include <unistd.h>  
#include <sys/wait.h>  
  
int main()  
{  
    int fd[2];  
  
    pipe(fd);  
  
    if (fork() == 0)  
    {  
        dup2(fd[1], STDOUT_FILENO);  
        close(fd[0]);  
        close(fd[1]);  
  
        execlp("ls", "ls", "-l", NULL);  
    }  
  
    if (fork() == 0)  
    {  
        dup2(fd[0], STDIN_FILENO);  
        close(fd[0]);  
        close(fd[1]);  
  
        execlp("grep", "grep", ".c", NULL);  
    }  
  
    close(fd[0]);  
    close(fd[1]);  
  
    wait(NULL);  
    wait(NULL);  
  
    return 0;  
}
```

```
sri_dhruthi@Dhruthi:~$ nano practical6b.c  
sri_dhruthi@Dhruthi:~$ gcc practical6b.c -o practical6b  
sri_dhruthi@Dhruthi:~$ ./practical6b  
-rwxr-xr-x 1 sri_dhruthi sri_dhruthi 16000 Aug 16 10:54 ExecExample  
-rw-r--r-- 1 sri_dhruthi sri_dhruthi 181 Aug 16 11:00 ExecExample.c  
-rwxr-xr-x 1 sri_dhruthi sri_dhruthi 16056 Aug 16 10:52 MultipleChildProcess  
-rw-r--r-- 1 sri_dhruthi sri_dhruthi 238 Aug 16 11:00 MultipleChildProcess  
.c  
-rwxr-xr-x 1 sri_dhruthi sri_dhruthi 16056 Aug 14 10:08 Multiplechildprocess  
-rw-r--r-- 1 sri_dhruthi sri_dhruthi 258 Aug 14 10:06 Multiplechildprocess  
.c  
-rw-r--r-- 1 sri_dhruthi sri_dhruthi 297 Aug 16 10:58 ParentChild.c  
-rw-r--r-- 1 sri_dhruthi sri_dhruthi 336 Aug 16 10:59 ParentWait.c  
-rwxr-xr-x 1 sri_dhruthi sri_dhruthi 16000 Aug 16 10:53 ProcessID  
-rw-r--r-- 1 sri_dhruthi sri_dhruthi 114 Aug 16 10:57 ProcessID.c  
-rw-r--r-- 1 sri_dhruthi sri_dhruthi 578 Sep 18 03:44 adresstranslating.c  
-rw-r--r-- 1 sri_dhruthi sri_dhruthi 79 Sep 28 15:24 combined.txt
```

2. Create a POSIX signal handling program that captures SIGINT, SIGTERM, and SIGUSR1. Demonstrate asynchronous event handling and explain the role of signal handlers.

```
#include <stdio.h>
#include <signal.h>
#include <unistd.h>

void handler(int sig)
{
    if(sig==SIGINT)
        printf("\nSIGINT Received\n");

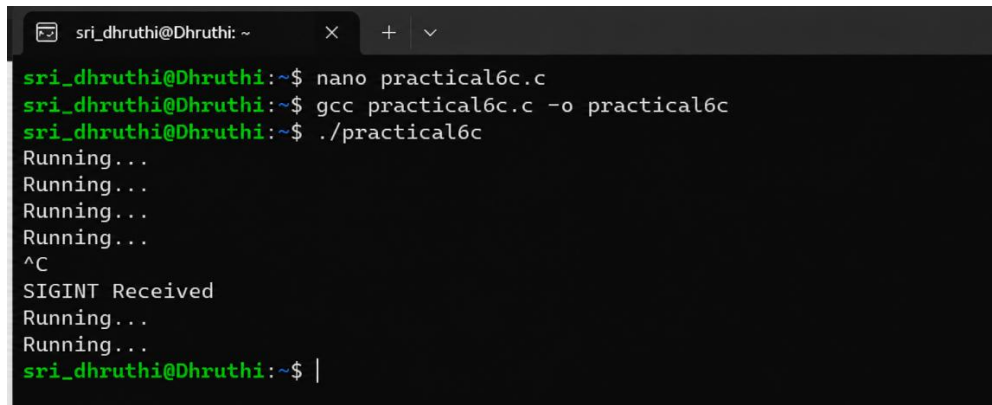
    if(sig==SIGTERM)
        printf("\nSIGTERM Received\n");

    if(sig==SIGUSR1)
        printf("\nSIGUSR1 Received\n");
}

int main()
{
    signal(SIGINT,handler);
    signal(SIGTERM,handler);
    signal(SIGUSR1,handler);

    while(1)
    {
        printf("Running...\n");
        sleep(3);
    }

    return 0;
}
```



```
sri_dhruthi@Dhruthi: ~
sri_dhruthi@Dhruthi:~$ nano practical6c.c
sri_dhruthi@Dhruthi:~$ gcc practical6c.c -o practical6c
sri_dhruthi@Dhruthi:~$ ./practical6c
Running...
Running...
Running...
Running...
^C
SIGINT Received
Running...
Running...
sri_dhruthi@Dhruthi:~$ |
```

Ex; kill -SIGUSR1 PID

Ex: kill -SIGTERM PID